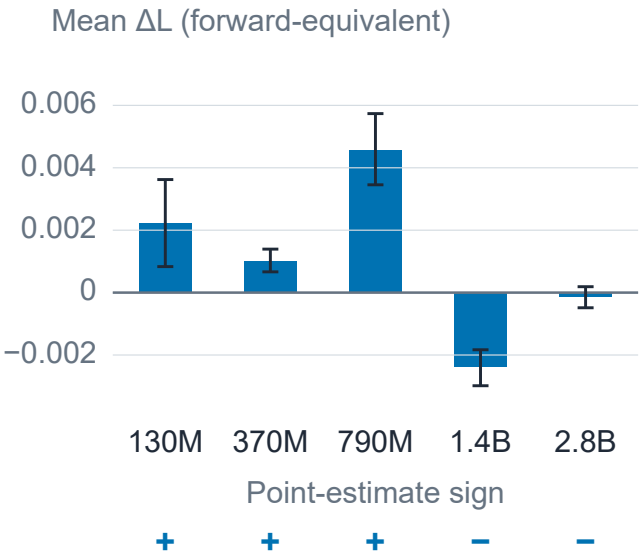
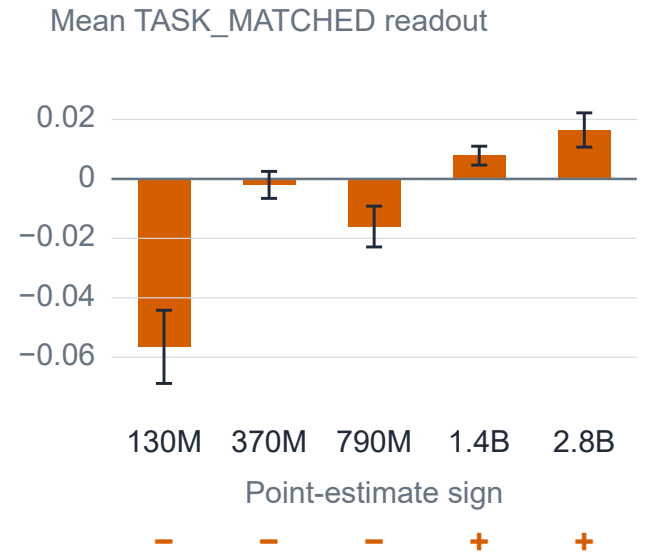


A Downstream task objective



B Vanilla next-token LM objective



Separate y-scales; bars and signs are point means, whiskers are percentile 95% pair intervals.

C Pair-level associations remain mixed

	130M	370M	790M	1.4B	2.8B
Pearson	+0.3208	-0.4593	+0.1862	-0.3664	+0.1330
Spearman	+0.2044	+0.0346	-0.0379	-0.1827	+0.0966
Pair sign agreement	0.5567	0.6033	0.4433	0.4567	0.5733

All five scales are shown descriptively; the 130M LM cell is the later completeness extension (Appendix A.6).